

AP[®] Calculus AB

Practice Exam

The questions contained in this AP[®] Calculus AB Practice Exam are written to the content specifications of AP Exams for this subject. Taking this practice exam should provide students with an idea of their general areas of strengths and weaknesses in preparing for the actual AP Exam. Because this AP Calculus AB Practice Exam has never been administered as an operational AP Exam, statistical data are not available for calculating potential raw scores or conversions into AP grades.

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A A

CALCULUS AB

SECTION I, Part A

Time—55 minutes

Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and place the letter of your choice in the corresponding box on the student answer sheet. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

GO ON TO THE NEXT PAGE.

1. $\int \cos(3x) \, dx =$

(B) $-\frac{1}{3}\sin(3x) + C$

(C) $\frac{1}{3}\sin(3x) + C$

(D) $\sin(3x) + C$

(E) $3\sin(3x) + C$

2. $\lim_{x \rightarrow 0} \frac{2x^6 + 6x^3}{4x^5 + 3x^3}$ is

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) nonexistent

-3-

13. $\int (x^3 + 1)^2 dx =$

(A) $\frac{1}{7}x^7 + x + C$

(B) $\frac{1}{7}x^7 + \frac{1}{2}x^4 + x + C$

(C) $6x^2(x^3 + 1) + C$

(D) $\frac{1}{3}(x^3 + 1)^3 + C$

(E) $\frac{(x^3 + 1)^3}{9x^2} + C$

14. $\lim_{h \rightarrow 0} \frac{e^{(2+h)} - e^2}{h} =$

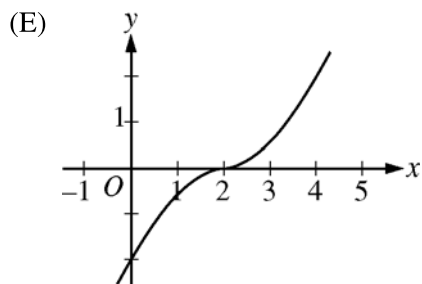
(A) 0 (B) 1 (C) $2e$ (D) e^2 (E) $2e^2$

-9-

- $$(E) \quad y = \frac{1}{1+x^2}$$

-10-

16. If $f'(x) = |x - 2|$, which of the following could be the graph of $y = f(x)$?



A graph of a function $y = f(x)$ on the interval $[0, 10]$. The function starts at the origin $(0, 0)$, reaches a local maximum at $x = 2$ with $y = 20$, crosses the x -axis at $x = 6$, reaches a local minimum at $x = 8$ with $y = 4$, and ends at $(10, 0)$.

21. The graph of f' , the derivative of the function f , is shown above for $0 \leq x \leq 10$. The areas of the regions between the graph of f' and the x -axis are 20, 6, and 4, respectively. If $f(0) = 2$, what is the maximum value of f on the closed interval $0 \leq x \leq 10$?

22. If $f'(x) = (x-2)(x-3)^2(x-4)^3$, then f has which of the following relative extrema?

- (A) I only
(B) III only
(C) I and III only
(D) II and III only
(E) I, II, and III

-14-

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B**B****B****B****B****B****B****B****B****CALCULUS AB****SECTION I, Part B****Time—50 minutes****Number of questions—17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and place the letter of your choice in the corresponding box on the student answer sheet. Do not spend too much time on any one problem.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

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B**B****B****B****B****B****B****B****B**

76. A particle moves along the x -axis so that at any time $t \geq 0$ its velocity is given by $v(t) = t^2 \ln(t + 2)$. What is the acceleration of the particle at time $t = 6$?

- (A) 1.500 (B) 20.453 (C) 29.453 (D) 74.860 (E) 133.417

77. If $\int_0^3 f(x) \, dx = 6$ and $\int_3^5 f(x) \, dx = 4$, then $\int_0^5 (3 + 2f(x)) \, dx =$

- (A) 10 (B) 20 (C) 23 (D) 35 (E) 50

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B**B****B****B****B****B****B****B****B**

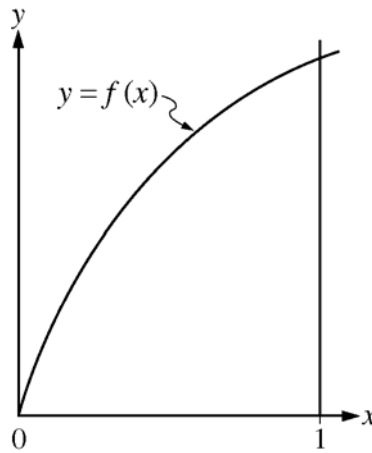
78. For $t \geq 0$ hours, H is a differentiable function of t that gives the temperature, in degrees Celsius, at an Arctic weather station. Which of the following is the best interpretation of $H'(24)$?

- (A) The change in temperature during the first day
- (B) The change in temperature during the 24th hour
- (C) The average rate at which the temperature changed during the 24th hour
- (D) The rate at which the temperature is changing during the first day
- (E) The rate at which the temperature is changing at the end of the 24th hour

79. A spherical tank contains 81.637 gallons of water at time $t = 0$ minutes. For the next 6 minutes, water flows out of the tank at the rate of $9\sin(\sqrt{t+1})$ gallons per minute. How many gallons of water are in the tank at the end of the 6 minutes?

- (A) 36.606 (B) 45.031 (C) 68.858 (D) 77.355 (E) 126.668

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B**B****B****B****B****B****B****B****B**

80. A left Riemann sum, a right Riemann sum, and a trapezoidal sum are used to approximate the value of $\int_0^1 f(x) \, dx$, each using the same number of subintervals. The graph of the function f is shown in the figure above. Which of the sums give an underestimate of the value of $\int_0^1 f(x) \, dx$?

- I. Left sum
 - II. Right sum
 - III. Trapezoidal sum
- (A) I only
(B) II only
(C) III only
(D) I and III only
(E) II and III only

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B**B****B****B****B****B****B****B****B**

81. The first derivative of the function f is given by $f'(x) = x - 4e^{-\sin(2x)}$. How many points of inflection does the graph of f have on the interval $0 < x < 2\pi$?

- (A) Three (B) Four (C) Five (D) Six (E) Seven

82. If f is a continuous function on the closed interval $[a, b]$, which of the following must be true?

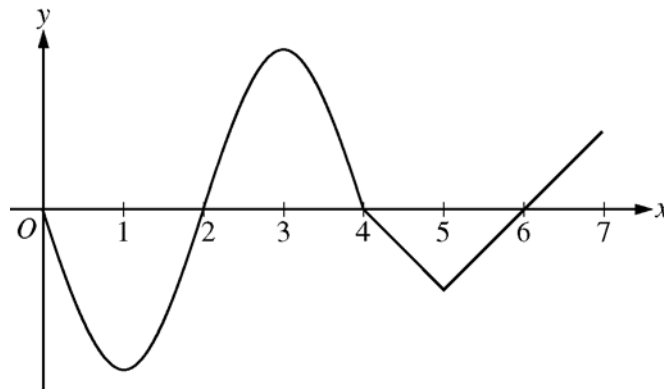
- (A) There is a number c in the open interval (a, b) such that $f(c) = 0$.
(B) There is a number c in the open interval (a, b) such that $f(a) < f(c) < f(b)$.
(C) There is a number c in the closed interval $[a, b]$ such that $f(c) \geq f(x)$ for all x in $[a, b]$.
(D) There is a number c in the open interval (a, b) such that $f'(c) = 0$.
(E) There is a number c in the open interval (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

B**B****B****B****B****B****B****B****B**

x	2.5	2.8	3.0	3.1
$f(x)$	31.25	39.20	45	48.05

83. The function f is differentiable and has values as shown in the table above. Both f and f' are strictly increasing on the interval $0 \leq x \leq 5$. Which of the following could be the value of $f'(3)$?

- (A) 20 (B) 27.5 (C) 29 (D) 30 (E) 30.5

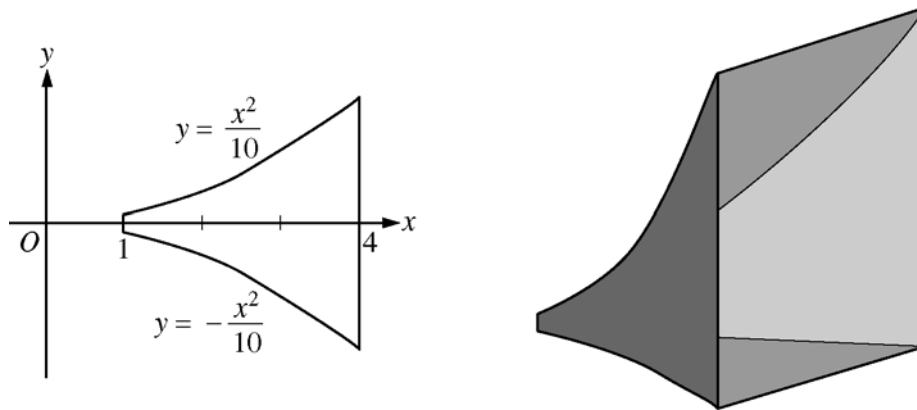


Graph of f'

84. The graph of f' , the derivative of the function f , is shown above. On which of the following intervals is f decreasing?

- (A) $[2, 4]$ only
(B) $[3, 5]$ only
(C) $[0, 1]$ and $[3, 5]$
(D) $[2, 4]$ and $[6, 7]$
(E) $[0, 2]$ and $[4, 6]$

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B**B****B****B****B****B****B****B****B**

85. The base of a loudspeaker is determined by the two curves $y = \frac{x^2}{10}$ and $y = -\frac{x^2}{10}$ for $1 \leq x \leq 4$, as shown in the figure above. For this loudspeaker, the cross sections perpendicular to the x -axis are squares. What is the volume of the loudspeaker, in cubic units?

(A) 2.046 (B) 4.092 (C) 4.200 (D) 8.184 (E) 25.711

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B**B****B****B****B****B****B****B****B**

x	3	4	5	6	7
$f(x)$	20	17	12	16	20

86. The function f is continuous and differentiable on the closed interval $[3, 7]$. The table above gives selected values of f on this interval. Which of the following statements must be true?

I. The minimum value of f on $[3, 7]$ is 12.

II. There exists c , for $3 < c < 7$, such that $f'(c) = 0$.

III. $f'(x) > 0$ for $5 < x < 7$.

(A) I only

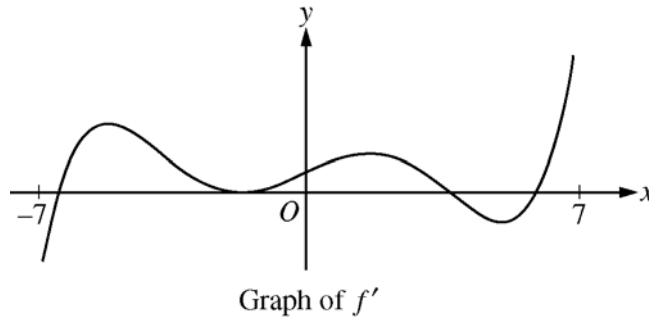
(B) II only

(C) III only

(D) I and III only

(E) I, II, and III

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B**B****B****B****B****B****B****B****B**

87. The figure above shows the graph of f' , the derivative of the function f , on the open interval $-7 < x < 7$. If f' has four zeros on $-7 < x < 7$, how many relative maxima does f have on $-7 < x < 7$?
- (A) One (B) Two (C) Three (D) Four (E) Five

-
88. The rate at which water is sprayed on a field of vegetables is given by $R(t) = 2\sqrt{1 + 5t^3}$, where t is in minutes and $R(t)$ is in gallons per minute. During the time interval $0 \leq t \leq 4$, what is the average rate of water flow, in gallons per minute?
- (A) 8.458 (B) 13.395 (C) 14.691 (D) 18.916 (E) 35.833

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B**B****B****B****B****B****B****B****B**

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-2	-3	4

89. The table above gives values of the differentiable functions f and g and their derivatives at $x = 1$. If $h(x) = (2f(x) + 3)(1 + g(x))$, then $h'(1) =$

- (A) -28 (B) -16 (C) 40 (D) 44 (E) 47

90. The functions f and g are differentiable. For all x , $f(g(x)) = x$ and $g(f(x)) = x$.

If $f(3) = 8$ and $f'(3) = 9$, what are the values of $g(8)$ and $g'(8)$?

- (A) $g(8) = \frac{1}{3}$ and $g'(8) = -\frac{1}{9}$
(B) $g(8) = \frac{1}{3}$ and $g'(8) = \frac{1}{9}$
(C) $g(8) = 3$ and $g'(8) = -9$
(D) $g(8) = 3$ and $g'(8) = -\frac{1}{9}$
(E) $g(8) = 3$ and $g'(8) = \frac{1}{9}$

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B**B****B****B****B****B****B****B****B**

91. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 5te^{-t} - 1$. At $t = 0$, the particle is at position $x = 1$. What is the total distance traveled by the particle from $t = 0$ to $t = 4$?
- (A) 0.366 (B) 0.542 (C) 1.542 (D) 1.821 (E) 2.821

-
92. Let f be the function with first derivative defined by $f'(x) = \sin(x^3)$ for $0 \leq x \leq 2$. At what value of x does f attain its maximum value on the closed interval $0 \leq x \leq 2$?
- (A) 0 (B) 1.162 (C) 1.465 (D) 1.845 (E) 2
-

END OF SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY
CHECK YOUR WORK ON PART B ONLY.**

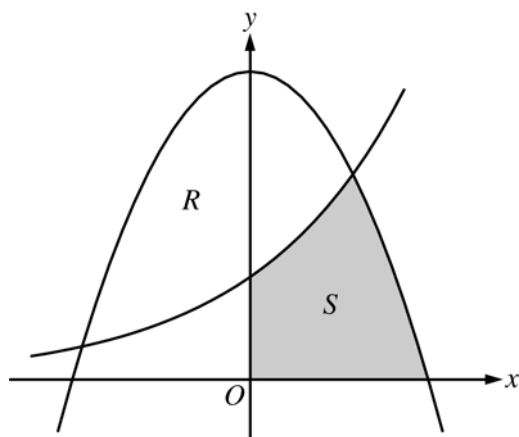
DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

CALCULUS AB
SECTION II, Part A
Time—45 minutes
Number of problems—3

A graphing calculator is required for some problems or parts of problems.

1. The rate at which raw sewage enters a treatment tank is given by $E(t) = 850 + 715 \cos\left(\frac{\pi t^2}{9}\right)$ gallons per hour for $0 \leq t \leq 4$ hours. Treated sewage is removed from the tank at the constant rate of 645 gallons per hour. The treatment tank is empty at time $t = 0$.
- (a) How many gallons of sewage enter the treatment tank during the time interval $0 \leq t \leq 4$? Round your answer to the nearest gallon.
- (b) For $0 \leq t \leq 4$, at what time t is the amount of sewage in the treatment tank greatest? To the nearest gallon, what is the maximum amount of sewage in the tank? Justify your answers.
- (c) For $0 \leq t \leq 4$, the cost of treating the raw sewage that enters the tank at time t is $(0.15 - 0.02t)$ dollars per gallon. To the nearest dollar, what is the total cost of treating all the sewage that enters the tank during the time interval $0 \leq t \leq 4$?

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2. Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.
- Find the area of S .
 - Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.
 - The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

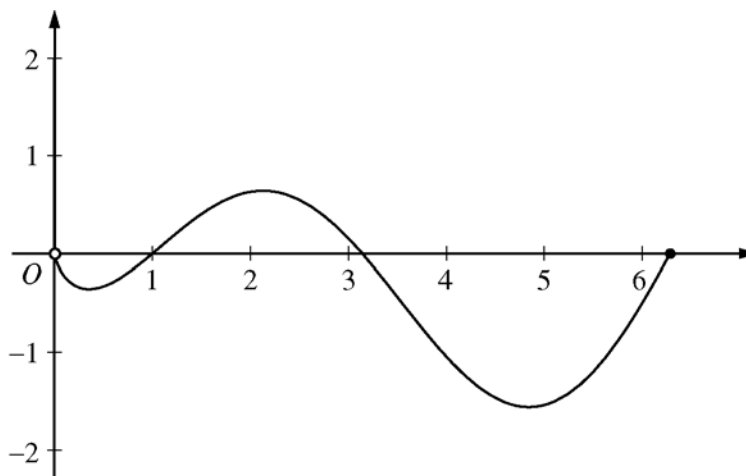
t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	90

3. The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.
- Use the data in the table to estimate the instantaneous rate at which the temperature of the oven is changing at time $t = 10$. Show the computations that lead to your answer. Indicate units of measure.
 - Write an integral expression in terms of H for the average temperature of the oven between time $t = 0$ and time $t = 16$. Estimate the average temperature of the oven using a left Riemann sum with four subintervals of equal length. Show the computations that lead to your answer.
 - Is your approximation in part (b) an underestimate or an overestimate of the average temperature? Give a reason for your answer.
 - Are the data in the table consistent with or do they contradict the claim that the temperature of the oven is increasing at an increasing rate? Give a reason for your answer.

END OF PART A OF SECTION II

CALCULUS AB
SECTION II, Part B
Time—45 minutes
Number of problems—3

No calculator is allowed for these problems.



Graph of f

4. Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$.

The function g is defined by $g(x) = \int_1^x f(t) \, dt$ for $0 < x \leq 2\pi$.

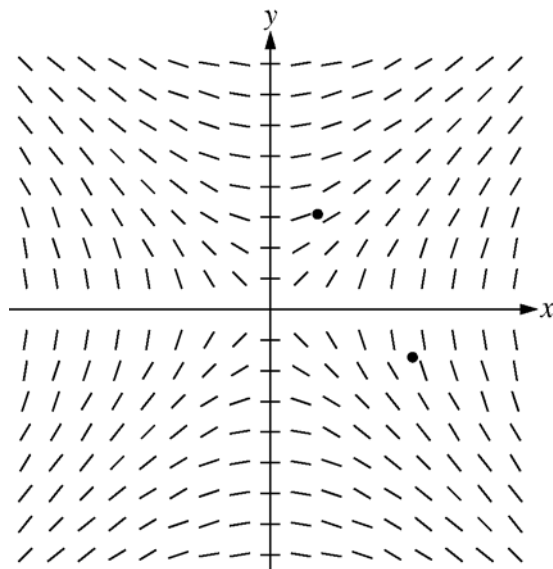
- (a) Find $g(1)$ and $g'(1)$.
- (b) On what intervals, if any, is g increasing? Justify your answer.
- (c) For $0 < x \leq 2\pi$, find the value of x at which g has an absolute minimum. Justify your answer.
- (d) For $0 < x < 2\pi$, is there a value of x at which the graph of g is tangent to the x -axis? Explain why or why not.

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5. Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, where $y \neq 0$.

- (a) The slope field for the given differential equation is shown below. Sketch the solution curve that passes through the point $(3, -1)$, and sketch the solution curve that passes through the point $(1, 2)$.

(Note: The points $(3, -1)$ and $(1, 2)$ are indicated in the figure.)



- (b) Write an equation for the line tangent to the solution curve that passes through the point $(1, 2)$.
- (c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(3) = -1$, and state its domain.

6. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

- (a) Find $\lim_{x \rightarrow \infty} g(x)$.
- (b) For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.
- (c) Find all values of b , if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.

STOP

END OF EXAM

AP[®] Calculus AB
Multiple-Choice Answer Key

No.	Correct Answer
1	C
2	D
3	C
4	B
5	C
6	B
7	A
8	D
9	C
10	D
11	B
12	D
13	B
14	D
15	E
16	E
17	D
18	D
19	A
20	C
21	C
22	A
23	B
24	D
25	A
26	C
27	E
28	E

No.	Correct Answer
76	C
77	D
78	E
79	A
80	D
81	B
82	C
83	D
84	E
85	D
86	B
87	A
88	C
89	D
90	E
91	D
92	C

AP[®] Calculus AB

Free-Response Scoring Guidelines

Question 1

The rate at which raw sewage enters a treatment tank is given by $E(t) = 850 + 715\cos\left(\frac{\pi t^2}{9}\right)$ gallons per hour for $0 \leq t \leq 4$ hours. Treated sewage is removed from the tank at the constant rate of 645 gallons per hour. The treatment tank is empty at time $t = 0$.

- (a) How many gallons of sewage enter the treatment tank during the time interval $0 \leq t \leq 4$? Round your answer to the nearest gallon.
- (b) For $0 \leq t \leq 4$, at what time t is the amount of sewage in the treatment tank greatest? To the nearest gallon, what is the maximum amount of sewage in the tank? Justify your answers.
- (c) For $0 \leq t \leq 4$, the cost of treating the raw sewage that enters the tank at time t is $(0.15 - 0.02t)$ dollars per gallon. To the nearest dollar, what is the total cost of treating all the sewage that enters the tank during the time interval $0 \leq t \leq 4$?

(a) $\int_0^4 E(t) dt \approx 3981$ gallons

- (b) Let $S(t)$ be the amount of sewage in the treatment tank at time t . Then $S'(t) = E(t) - 645$ and $S'(t) = 0$ when $E(t) = 645$. On the interval $0 \leq t \leq 4$, $E(t) = 645$ when $t = 2.309$ and $t = 3.559$.

t (hours)	amount of sewage in treatment tank
0	0
2.309	$\int_0^{2.309} E(t) dt - 645(2.309) = 1637.178$
3.559	$\int_0^{3.559} E(t) dt - 645(3.559) = 1228.520$
4	$3981.022 - 645(4) = 1401.022$

The amount of sewage in the treatment tank is greatest at $t = 2.309$ hours. At that time, the amount of sewage in the tank, rounded to the nearest gallon, is 1637 gallons.

(c) Total cost $= \int_0^4 (0.15 - 0.02t)E(t) dt = 474.320$

The total cost of treating the sewage that enters the tank during the time interval $0 \leq t \leq 4$, to the nearest dollar, is \$474.

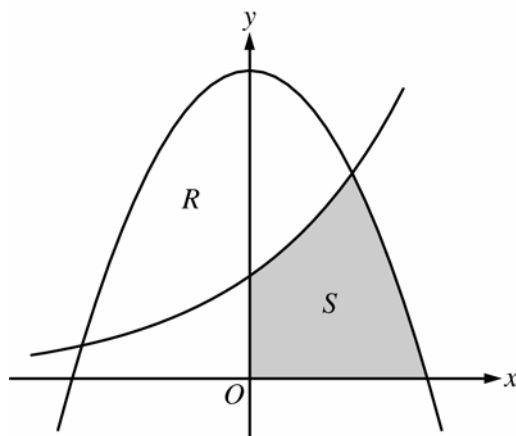
2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

4 : $\begin{cases} 1 : \text{sets } E(t) = 645 \\ 1 : \text{identifies } t = 2.309 \text{ as a candidate} \\ 1 : \text{amount of sewage at } t = 2.309 \\ 1 : \text{conclusion} \end{cases}$

3 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{limits} \\ 1 : \text{answer} \end{cases}$

AP[®] Calculus AB
Free-Response Scoring Guidelines

Question 2



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

- (a) Find the area of S .
- (b) Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.
- (c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

Let $a = -1.63658$

$$\begin{aligned} \text{(a) Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$

$$3 : \begin{cases} 1 : \text{integrand} \\ 1 : \text{limits} \\ 1 : \text{answer} \end{cases}$$

$$\begin{aligned} \text{(b) Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107 \end{aligned}$$

$$4 : \begin{cases} 2 : \text{integrand} \\ 1 : \text{limits and constant} \\ 1 : \text{answer} \end{cases}$$

$$\text{(c) Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

$$2 : \begin{cases} 1 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$$

AP[®] Calculus AB
Free-Response Scoring Guidelines

Question 3

t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	90

The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

- Use the data in the table to estimate the instantaneous rate at which the temperature of the oven is changing at time $t = 10$. Show the computations that lead to your answer. Indicate units of measure.
- Write an integral expression in terms of H for the average temperature of the oven between time $t = 0$ and time $t = 16$. Estimate the average temperature of the oven using a left Riemann sum with four subintervals of equal length. Show the computations that lead to your answer.
- Is your approximation in part (b) an underestimate or an overestimate of the average temperature? Give a reason for your answer.
- Are the data in the table consistent with or do they contradict the claim that the temperature of the oven is increasing at an increasing rate? Give a reason for your answer.

$$(a) \quad H'(10) \approx \frac{H(12) - H(8)}{12 - 8} = \frac{80 - 73}{4} = \frac{7}{4}^{\circ}\text{C}/\text{min}$$

$$2 : \begin{cases} 1 : \text{difference quotient} \\ 1 : \text{answer with units} \end{cases}$$

$$(b) \quad \text{Average temperature is } \frac{1}{16} \int_0^{16} H(t) \, dt$$

$$\int_0^{16} H(t) \, dt \approx 4 \cdot (65 + 68 + 73 + 80)$$

$$\text{Average temperature} \approx \frac{4 \cdot 286}{16} = 71.5^{\circ}\text{C}$$

$$3 : \begin{cases} 1 : \frac{1}{16} \int_0^{16} H(t) \, dt \\ 1 : \text{left Riemann sum} \\ 1 : \text{answer} \end{cases}$$

- The left Riemann sum approximation is an underestimate of the integral because the graph of H is increasing. Dividing by 16 will not change the inequality, so 71.5°C is an underestimate of the average temperature.

$$1 : \text{answer with reason}$$

- If a continuous function is increasing at an increasing rate, then the slopes of the secant lines of the graph of the function are increasing. The slopes of the secant lines for the four intervals in the table are $\frac{3}{4}$, $\frac{5}{4}$, $\frac{7}{4}$, and $\frac{10}{4}$, respectively.

Since the slopes are increasing, the data are consistent with the claim.

OR

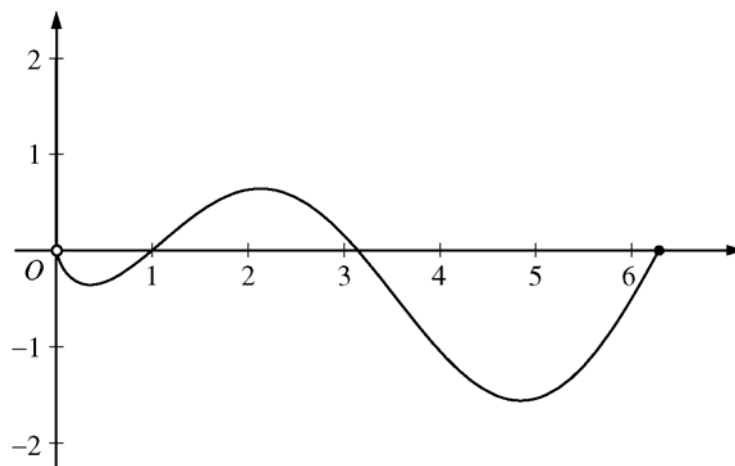
By the Mean Value Theorem, the slopes are also the values of $H'(c_k)$ for some times $c_1 < c_2 < c_3 < c_4$, respectively.

Since these derivative values are positive and increasing, the data are consistent with the claim.

$$3 : \begin{cases} 1 : \text{considers slopes of four secant lines} \\ 1 : \text{explanation} \\ 1 : \text{conclusion consistent with explanation} \end{cases}$$

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Question 4



Graph of f

Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$. The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

- Find $g(1)$ and $g'(1)$.
- On what intervals, if any, is g increasing? Justify your answer.
- For $0 < x \leq 2\pi$, find the value of x at which g has an absolute minimum. Justify your answer.
- For $0 < x < 2\pi$, is there a value of x at which the graph of g is tangent to the x -axis? Explain why or why not.

(a) $g(1) = \int_1^1 f(t) dt = 0$ and $g'(1) = f(1) = 0$

2 : $\begin{cases} 1 : g(1) \\ 1 : g'(1) \end{cases}$

(b) Since $g'(x) = f(x)$, g is increasing on the interval $1 \leq x \leq \pi$ because $f(x) > 0$ for $1 < x < \pi$.

2 : $\begin{cases} 1 : \text{interval} \\ 1 : \text{reason} \end{cases}$

(c) For $0 < x < 2\pi$, $g'(x) = f(x) = 0$ when $x = 1, \pi$. $g' = f$ changes from negative to positive only at $x = 1$. The absolute minimum must occur at $x = 1$ or at the right endpoint. Since $g(1) = 0$ and $g(2\pi) = \int_1^{2\pi} f(t) dt = \int_1^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt < 0$ by comparison of the two areas, the absolute minimum occurs at $x = 2\pi$.

3 : $\begin{cases} 1 : \text{identifies 1 and } 2\pi \text{ as candidates} \\ \quad \text{- or -} \\ \quad \text{indicates that the graph of } g \\ \quad \text{decreases, increases, then decreases} \\ 1 : \text{justifies } g(2\pi) < g(1) \\ 1 : \text{answer} \end{cases}$

(d) Yes, the graph of g is tangent to the x -axis at $x = 1$ since $g(1) = 0$ and $g'(1) = 0$.

2 : $\begin{cases} 1 : \text{answer of "yes" with } x = 1 \\ 1 : \text{explanation} \end{cases}$

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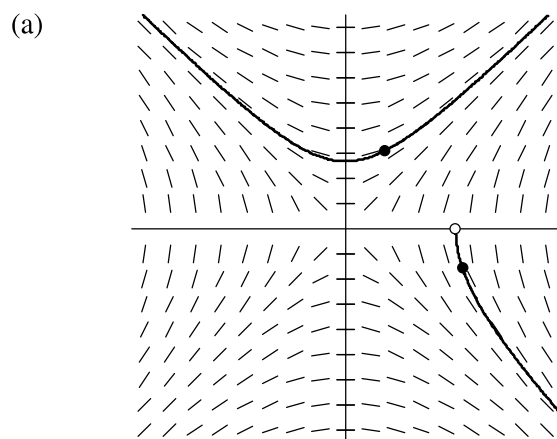
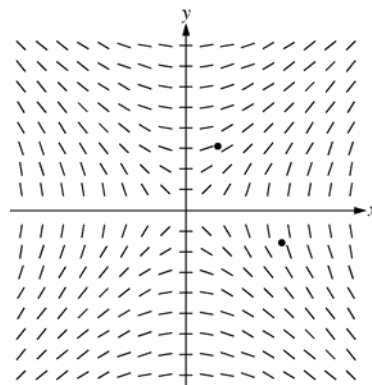
Question 5

Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, where $y \neq 0$.

- (a) The slope field for the given differential equation is shown below. Sketch the solution curve that passes through the point $(3, -1)$, and sketch the solution curve that passes through the point $(1, 2)$.

(Note: The points $(3, -1)$ and $(1, 2)$ are indicated in the figure.)

- (b) Write an equation for the line tangent to the solution curve that passes through the point $(1, 2)$.
- (c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(3) = -1$, and state its domain.



(b) $\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{1}{2}$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.

(c) $y \, dy = x \, dx$
 $\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$
 $y^2 = x^2 + C$
 $C = -8$

Since the particular solution goes through $(3, -1)$, y must be negative.

$$y = -\sqrt{x^2 - 8} \text{ for } x > \sqrt{8}$$

2 : $\begin{cases} 1 : \text{solution curve through } (3, -1) \\ 1 : \text{solution curve through } (1, 2) \end{cases}$

Curves must go through the indicated points, follow the given slope lines, and extend to the boundary of the slope field or the x -axis.

1 : equation of tangent line

6 : $\begin{cases} 1 : \text{separates variables} \\ 1 : \text{antiderivatives} \\ 5 : \begin{cases} 1 : \text{constant of integration} \\ 1 : \text{uses initial condition} \\ 1 : \text{solves for } y \end{cases} \\ \text{Note: max } 2/5 \text{ [1-1-0-0-0] if no constant of integration} \\ 1 : \text{domain} \end{cases}$

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Question 6

Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

- (a) Find $\lim_{x \rightarrow \infty} g(x)$.
- (b) For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.
- (c) Find all values of b , if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.

(a) $\lim_{x \rightarrow \infty} g(x) = 0$

1 : answer

(b) $g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$

$$g'\left(\frac{2}{3}\right) = \left(\frac{1}{3} - b\right)e^{-2/3} = 0 \Rightarrow b = \frac{1}{3}$$

When $b = \frac{1}{3}$, $g'(x) = \left(\frac{2}{3} - x\right)e^{-x}$.

For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum

at $x = \frac{2}{3}$.

$$4 : \begin{cases} 2 : g'(x) \\ 1 : \text{solves } g'\left(\frac{2}{3}\right) = 0 \text{ for } b \\ 1 : \text{justification} \end{cases}$$

(c) $g''(x) = -e^{-x} - (1 - x - b)e^{-x} = (x - 2 + b)e^{-x}$

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.

Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

$$4 : \begin{cases} 2 : g''(x) \\ 1 : \text{interval for } b \\ 1 : \text{justification} \end{cases}$$